

Assignment 4

Understanding of Conditional Instructions in Java

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Objective

The objective of this assignment is to understand and implement conditional instructions (if, if-else, and nested if-else) in Java by solving a variety of real-world decision-making problems.

1. Check Even/Odd and Positive/Negative

This program reads a number from the user and determines whether it is positive, negative, or zero, and also checks whether it is even or odd using if-else statements.

Java Program

```
import java.util.Scanner;

public class EvenOddPosNeg {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        // Check Positive / Negative / Zero
        if (num > 0) {
            System.out.println(num + " is Positive");
        } else if (num < 0) {
            System.out.println(num + " is Negative");
        } else {
            System.out.println(num + " is Zero");
        }

        // Check Even / Odd (applicable for non-zero numbers as well)
        if (num % 2 == 0) {
            System.out.println(num + " is Even");
        } else {
            System.out.println(num + " is Odd");
        }

        sc.close();
    }
}
```

```
Enter a number: 12
12 is Positive
12 is Even

$ java EvenOddPosNeg.java
Enter a number: -7
-7 is Negative
-7 is Odd
```

Output

2. Largest of Three Numbers

This program reads three numbers from the user and determines the largest among them using if-else statements.

Java Program

```
import java.util.Scanner;

public class LargestOfThree {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();
        System.out.print("Enter second number: ");
        int b = sc.nextInt();
        System.out.print("Enter third number: ");
        int c = sc.nextInt();

        int largest;
        if (a >= b && a >= c) {
            largest = a;
        } else if (b >= a && b >= c) {
            largest = b;
        } else {
            largest = c;
        }

        System.out.println("The largest number is: " + largest);

        sc.close();
    }
}
```

Output

```
Enter first number: 15
Enter second number: 42
Enter third number: 8
The largest number is: 42
```

3. Number Divisibility Checker

This program reads a number and checks whether it is divisible by 5. If it is divisible by 5, a nested if-else further checks whether it is also divisible by 10.

Java Program

```
import java.util.Scanner;

public class DivisibilityChecker {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        if (num % 5 == 0) {
            System.out.println(num + " is divisible by 5");

            if (num % 10 == 0) {
                System.out.println(num + " is also divisible by 10");
            } else {
                System.out.println(num + " is not divisible by 10");
            }
        } else {
            System.out.println(num + " is not divisible by 5");
        }

        sc.close();
    }
}
```

Output

```
Enter a number: 50
50 is divisible by 5
50 is also divisible by 10

$ java DivisibilityChecker.java
Enter a number: 35
35 is divisible by 5
35 is not divisible by 10
```

4. Check Working Day

This program reads a day number from 1 to 7 (1 = Monday, 7 = Sunday) and determines whether it is a working day (Monday to Friday) or a weekend (Saturday or Sunday).

Java Program

```
import java.util.Scanner;

public class WorkingDayChecker {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter day number (1-7, where 1 = Monday and 7 = Sunday): ");
        int day = sc.nextInt();

        if (day >= 1 && day <= 5) {
            System.out.println("Day " + day + " is a Working Day");
        } else if (day == 6 || day == 7) {
            System.out.println("Day " + day + " is a Weekend");
        } else {
            System.out.println("Invalid day number. Please enter a value between 1 and
7.");
        }

        sc.close();
    }
}
```

Output

```
Enter day number (1-7, where 1 = Monday and 7 = Sunday): 3
Day 3 is a Working Day

$ java WorkingDayChecker.java
Enter day number (1-7, where 1 = Monday and 7 = Sunday): 7
Day 7 is a Weekend
```

5. Password Length Checker

This program reads a password from the user and classifies its strength as Weak (less than 8 characters), Acceptable (8 to 11 characters), or Strong (12 or more characters) based on its length.

Java Program

```
import java.util.Scanner;

public class PasswordLengthChecker {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a password: ");
        String password = sc.nextLine();

        int length = password.length();

        if (length < 8) {
            System.out.println("Password Strength: Weak (less than 8 characters)");
        } else if (length >= 8 && length <= 11) {
            System.out.println("Password Strength: Acceptable (8 to 11 characters)");
        } else {
            System.out.println("Password Strength: Strong (12 or more characters)");
        }

        sc.close();
    }
}
```

Output

```
Enter a password: abc123
Password Strength: Weak (less than 8 characters)

$ java PasswordLengthChecker.java
Enter a password: MyStrongPass123
Password Strength: Strong (12 or more characters)
```